CHAPTER 6: RISK AVERSION AND CAPITAL ALLOCATION TO RISKY ASSETS

1. (d) While a higher or lower trading costs are not necessarily an indication of an investor's tolerance for risk, risk-averse investors may prefer the portfolios associated with lower trading costs (less actively managed, and therefore less risk). Investors with a higher degree of risk aversion will not want (a) higher risk premiums (as they are associated with higher risks), (b) riskier portfolios (higher standard deviation), or (c) lower Sharpe Ratios (any investor will always prefer investment portfolios with higher Sharpe ratios)

2. (b) A higher borrowing rate is a consequence of the risk of the borrowers' default. In perfect markets with no additional cost of default, this increment would equal the value of the borrower's option to default, and the Sharpe measure, with appropriate treatment of the default option, would be the same. However, there are costs to default so that this part of the increment lowers the Sharpe ratio. Also, notice that answer (c) is not correct because doubling the expected return with a fixed risk-free rate will more than double the risk premium and the Sharpe ratio.

3. Assuming no change in risk tolerance, that is, an unchanged risk-aversion coefficient (and A is generally greater than zero), higher perceived volatility increases the denominator of the equation for the optimal investment in the risky portfolio:

$$y\times = \frac{E(r_P) - r_f}{A\sigma_P^2} \qquad \text{(Equation 6.7).}$$

The proportion invested in the risky portfolio will therefore decrease.

4.

a. The expected cash flow is: (0.5 × \$70,000) + (0.5 × 200,000) = \$135,000. With a risk premium of 8% over the risk-free rate of 2%, the required rate of return is 10%. Therefore, the present value of the portfolio is:

$$\$135,000/1.10 = \$\mathbf{122,727}$$

b. If the portfolio is purchased for \$122,727 and provides an expected cash inflow of \$135,000, then the expected rate of return ($E([r]$) is as follows:

$\$122,727 \times (1 + E[r]) = \$135,000$

Therefore, $E(r)$ = **10%**. The portfolio price is set to equate the expected rate of return with the required rate of return.

c. If the risk premium over T-bills is now 12%, then the required return is:

2% + 12% = 14%

The present value of the portfolio is now: $\$135{,}000 / 1.14 = \$\mathbf{118{,}421}$

d. For a given expected cash flow, portfolios that command greater risk premiums must sell at lower prices. The extra discount from expected value is a penalty for risk.

5. When we specify utility by $U = E(r) - 0.5A\sigma^2$, the utility level for T-bills is: 0.02
The utility level for the risky portfolio is:

$$U = 0.07 - 0.5 \times A \times (0.18)^2 = 0.07 - 0.0162 \times A$$

In order for the risky portfolio to be preferred to T-bills, the following must hold:

$$0.07 - 0.0162 \times A > 0.02 \rightarrow A < 0.05 / 0.0162 = \mathbf{3.09}$$

A must be less than 3.09 for the risky portfolio to be preferred to T-bills.

6. Answers will vary. Points on the curve are derived by solving for *E*(*r*) in the following equation: $U = 0.02 = E(r) ?\ .5A\sigma^2 = E(r) - 0.5 \times 3 \times \sigma^2$

The values of *E*(*r*), given the values of σ^2, are therefore:

σ	σ^2	*E*(*r*)
0.00	0.0000	0.0200
0.05	0.0025	0.0238
0.10	0.0100	0.0350
0.15	0.0225	0.0538
0.20	0.0400	0.0800
0.25	0.0625	0.1138

The bold line in the graph on the next page (labeled Q6, for Question 6) depicts the indifference curve.

7. Repeating the analysis in Problem 6, utility is now:

$$U = E(r) - 0.5A\sigma^2 = E(r) - 0.5 \times 4 \times \sigma^2 = 0.02$$

The equal–utility combinations of expected return and standard deviation are presented in the table below. The indifference curve is the upward sloping line in the graph on the next page, labeled Q7 (for Question 7).

σ	σ^2	*E*(*r*)
0.00	0.0000	0.0200
0.05	0.0025	0.0250
0.10	0.0100	0.0400
0.15	0.0225	0.0650
0.20	0.0400	0.1000
0.25	0.0625	0.1450

The indifference curve in Problem 7 differs from that in Problem 6 in slope. When *A* increases from 3 to 4, the increased risk aversion results in a greater slope for the indifference curve since more expected return is needed to compensate for additional σ.

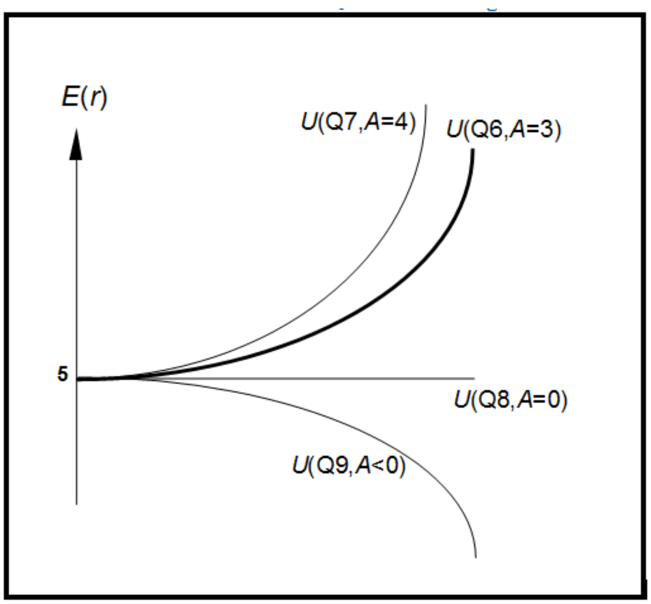
8. The coefficient of risk aversion for a risk neutral investor is zero. Therefore, the corresponding utility is equal to the portfolio's expected return. The corresponding indifference curve in the expected return–standard deviation plane is a horizontal line, labeled Q8 in the graph above (see Problem 6).

σ	σ^2	$E(r)$
0.00	0.0000	0.0200
0.05	0.0025	0.0200
0.10	0.0100	0.0200
0.15	0.0225	0.0200
0.20	0.0400	0.0200
0.25	0.0625	0.0200

9. A risk lover (here, A = −3), rather than penalizing portfolio utility to account for risk, derives greater utility as variance increases. This amounts to a negative coefficient of risk aversion. The corresponding indifference curve is downward sloping in the graph above (see Problem 6) and is labeled Q9.

σ	σ^2	$E(r)$
0.00	0.0000	0.0200
0.05	0.0025	0.0163
0.10	0.0100	0.0050
0.15	0.0225	(0.0138)
0.20	0.0400	(0.0400)
0.25	0.0625	(0.0738)

Indifference curves for Questions 6 through 9:



10. The portfolio expected return and variance are computed as follows:

(1) W_{Bills}	(2) r_{Bills}	(3) W_{Inex}	(4) r_{Index}	$r_{\text{Portfolio}}$ (1)×(2)+(3)×(4)	$\sigma_{\text{Portfolio}}$ (3) × 20%	$\sigma^2_{\text{Portfolio}}$
0.0	2%	1.0	10.0%	**10.00%**	20.00%	**0.0400**
0.2	2	0.8	10.0	**8.40%**	16.00%	**0.0256**
0.4	2	0.6	10.0	**6.80%**	12.00%	**0.0144**
0.6	2	0.4	10.0	**5.20%**	8.00%	**0.0064**
0.8	2	0.2	10.0	**3.60%**	4.00%	**0.0016**
1.0	2	0.0	10.0	**2.00%**	0.00%	**0.0000**

11. Computing utility from $U = E(r) - 0.5 \times A\sigma^2 = E(r) - \sigma^2$, we arrive at the values in the column labeled *U*(*A* = 2) in the following table:

W_{Bills}	W_{Inex}	$r_{\text{Portfolio}}$	$\sigma_{\text{Portfolio}}$	$\sigma^2_{\text{Portfolio}}$	*U*(*A* = 2)	*U*(*A* = 3)
0.0	1.0	10.00%	20.00%	0.0400	**0.0600**	0.0400
0.2	0.8	8.40%	16.00%	0.0256	0.0584	0.0456
0.4	0.6	6.80%	12.00%	0.0144	0.0536	**0.0464**
0.6	0.4	5.20%	8.00%	0.0064	0.0456	0.0424
0.8	0.2	3.60%	4.00%	0.0016	0.0344	0.0336
1.0	0.0	2.00%	0.00%	0.0000	0.0200	0.0200

The column labeled *U*(*A* = 2) implies that investors with *A* = 2 prefer a portfolio that is invested **100%** in the market index to any of the other portfolios in the table.

12. The column labeled *U*(*A* = 3) in the table above is computed from:

$$U = E(r) - 0.5A\sigma^2 = E(r) - 1.5\sigma^2$$

The more risk averse investors (A = 3) prefer the portfolio that is invested **60%** in the market, rather than the 100% market weight preferred by investors with *A* = 2.

13. Expected return = (0.7 × 12%) + (0.3 × 2%) = **9%**

Standard deviation = 0.7 × 28% = **19.6%**

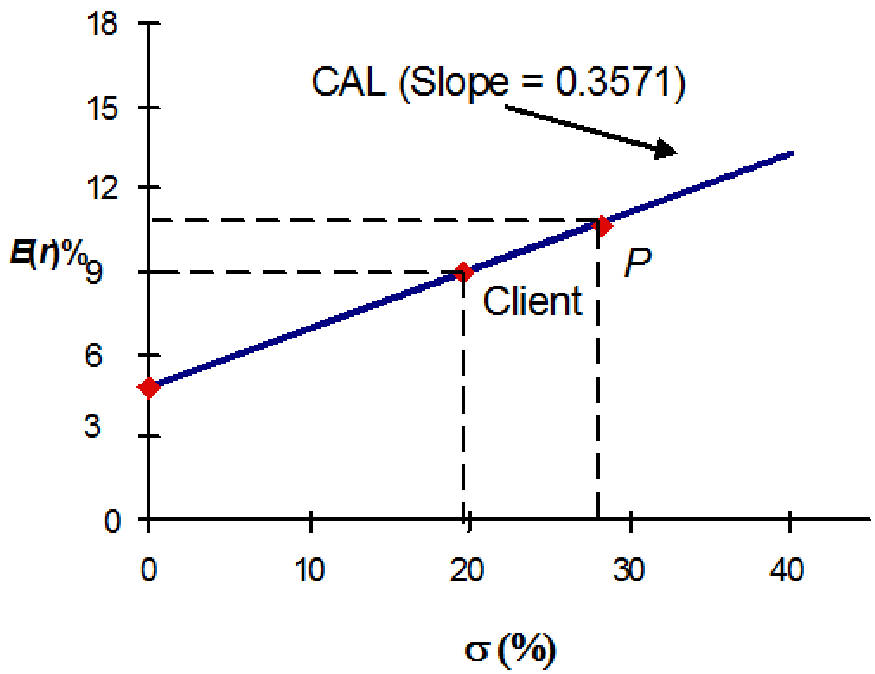
14.

Investment proportions: **30.0%** in T-bills
0.7 × 25% = **17.5%** in Stock A
0.7 × 32% = **22.4%** in Stock B
0.7 × 43% = **30.1%** in Stock C

15. Your reward-to-volatility (Sharpe) ratio: $S = \dfrac{0.12 - 0.02}{0.28} = \mathbf{0.3571}$

Client's reward-to-volatility (Sharpe) ratio: $S = \dfrac{0.09 - 0.02}{0.196} = \mathbf{0.3571}$

16.



17.

a. $E(r_C) = r_f + y \times (E[r_P] - r_f) = 0.02 + y \times (0.12 - 0.02)$

If the expected return for the portfolio is 16%, then:

$$10\% = 2\% + 10\% \times y \Rightarrow y = \frac{0.10 - 0.02}{0.10} = \mathbf{0.8}$$

Therefore, in order to have a portfolio with expected rate of return equal to 10%, the client must invest 80% of total funds in the risky portfolio and 20% in T-bills.

b.

Client's investment proportions:	**20.0%** in T-bills
	0.8 × 25% = **20.0%** in Stock A
	0.8 × 32% = **25.6%** in Stock B
	0.8 × 43% = **34.4%** in Stock C

c. $\sigma_C = 0.8 \times \sigma_P = 0.8 \times 28\% = \mathbf{22.4\%}$

18.

a. $\sigma_C = y \times 28\%$

If your client prefers a standard deviation of at most 12%, then:

$y = 0.12/0.28 = 0.4286 = \mathbf{42.86}\%$ invested in the risky portfolio.

b. $E(r_C) = 0.02 + 0.1 \times y = 0.02 + (0.1 \times 0.4286) = 0.0629 = \mathbf{6.29}\%$

19.

a. $y* = \frac{E(r_P)-r_f}{A\sigma_P^2} = \frac{0.12-0.02}{3.5\times 0.28^2} = \frac{0.10}{0.2744} = 0.3644$

Therefore, the client's optimal proportions are: **36.44%** invested in the risky portfolio and **63.56%** invested in T-bills.

b. $\boldsymbol{E(r_C) = 0.02 + 0.10 \times y = 0.02 + (0.3644 \times 0.1) = 0.05644\ or 5.644\%}$

$$\boldsymbol{\sigma_C = 0.3644 \times 28 = 10.203\%}$$

20.

a. If the period 1927–2021 is assumed to be representative of future expected performance, then we use the following data to compute the fraction allocated to equity: $A = 4, E(r_M) - r_f = 8.87\%,\ \sigma_M = 20.25\%$ (we use the standard deviation of the risk premium from Table 6.7). Then y^* is given by:

$$y^* = \frac{E(r_M)-r_f}{A\sigma_M^2} = \frac{0.0887}{4\times 0.2025^2} = 0.5408$$

That is, **54.08%** of the portfolio should be allocated to equity and **45.92%** should be allocated to T-bills.

b. If the period 1975–1998 is assumed to be representative of future expected performance, then we use the following data to compute the fraction allocated to equity: $A = 4, E(r_M) - r_f = 11.00\%,\ \sigma_M = 14.40\%$ and y^* is given by:

$$y^* = \frac{E(r_M)-r_f}{A\sigma_M^2} = \frac{.1100}{4\times .1440^2} = 1.3262$$

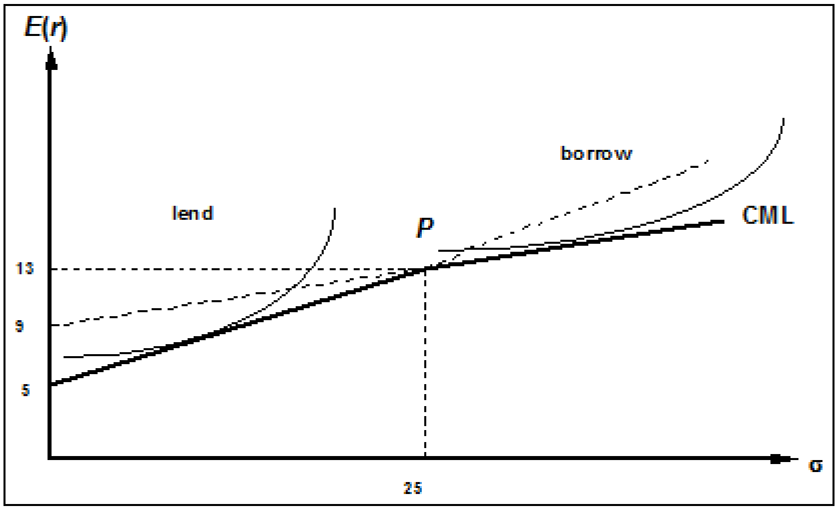
Therefore, **132.62%** of the complete portfolio should be allocated to equity by borrowing **32.62%** in T-bills.

c. In Part b, the market risk premium is expected to be higher than in Part a but market risk is lower than in Part a. Therefore, the reward-to-volatility *ratio* is expected to be much higher Part b, which explains borrowing against T-bills to expand exposure to equity.

23.

Data: $r_f = 2\%, E(r_M) = 10\%,\ \sigma_M = 25\%,\ \text{and}\ r_f^B = 6\%$

The CML and indifference curves are as follows:



24.

For y to be less than 1.0 (that the investor is a lender), risk aversion (A) must be large enough such that:

$$y = \frac{E(r_M) - r_f}{A\sigma_M^2} < 1 \Rightarrow A > \frac{0.10 - 0.02}{0.25^2} = \mathbf{1.28}$$

For y to be greater than 1 (the investor is a borrower), A must be small enough:

$$y = \frac{E(r_M) - r_f}{A\sigma_M^2} < 1 \Rightarrow A > \frac{0.10 - 0.06}{0.25^2} = \mathbf{0.64}$$

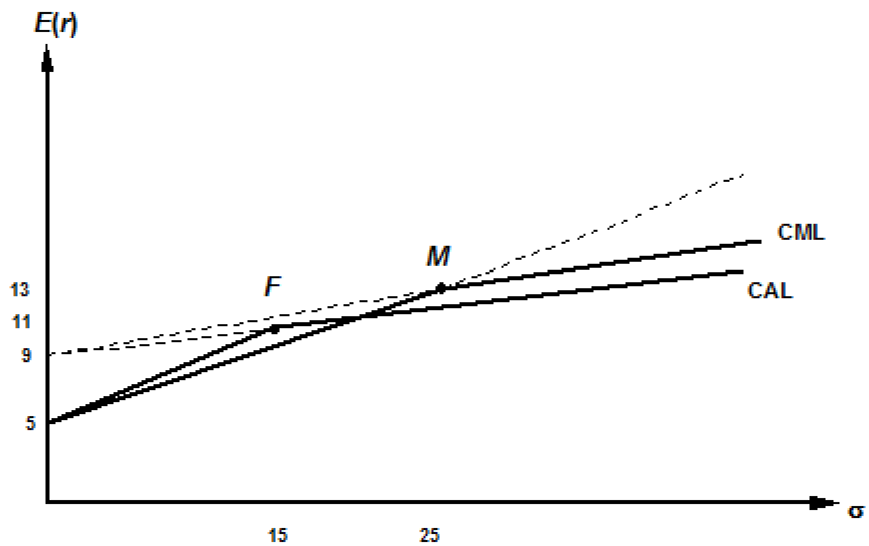
For values of risk aversion within this range, the client will neither borrow nor lend but will hold a portfolio composed only of the optimal risky portfolio:

$\boldsymbol{y = 1 \text{ for } 0.64 \le A \le 1.28}$

25.

a. The graph for Problem 23 must be redrawn here, with:

$E(r_P) = 8\%$ and $\sigma_P = 15\%$



b. For a lending position: $A > \dfrac{0.08 - 0.02}{0.15^2} = 2.67$

For a borrowing position: $A < \dfrac{0.08 - 0.06}{0.15^2} = 0.89$

Therefore, $y = 1$ for $.89 \le A \le 2.67$

26. The maximum feasible fee, denoted *f*, depends on the reward-to-variability ratio. For $y < 1$, the lending rate, 2%, is viewed as the relevant risk-free rate, and we solve for *f* as follows:

$$\frac{0.08 - 0.02 - f}{0.15} = \frac{0.10 - 0.02}{0.25} \Rightarrow f = 0.06 - \frac{0.15 \times 0.08}{0.25} = 0.012 \text{ or } \mathbf{1.2\%}$$

For $y > 1$, the borrowing rate, 6%, is the relevant risk-free rate. Then we notice that, even without a fee, the active fund is inferior to the passive fund because:

$$\frac{0.08 - 0.06 - f}{0.15} = \frac{0.10 - 0.06}{0.25} \rightarrow f = -0.004, \text{or} -\mathbf{.4}\%$$

More risk tolerant investors (who are more inclined to borrow) will not be clients of the fund. We find that *f* is negative: that is, you would need to *pay* investors to choose your active fund. These investors desire higher risk–higher return complete portfolios and thus are in the borrowing range of the relevant CAL. In this range, the reward-to-variability ratio of the index (the passive fund) is better than that of the managed fund.

CFA PROBLEMS

1. Utility for each investment $= E(r) - 0.5 \times 4 \times \sigma^2$

   We choose the investment with the highest utility value, Investment 3.

Investment	Expected return $E(r)$	Standard deviation σ	Utility U
1	0.12	0.30	−0.0600
2	0.15	0.50	−0.3500
3	0.21	0.16	0.1588
4	0.24	0.21	0.1518

2. When investors are risk neutral, then $A = 0$; the investment with the highest utility is Investment 4 because it has the highest expected return.

3. (b)

4. Indifference curve 2 because it is tangent to the CAL.

5. Point E

6. (0.6 × \$50,000) + (0.4 × [–\$30,000]) – \$5,000 = \$13,000

7. (b) Higher borrowing rates will reduce the total return to the portfolio and this results in a part of the line that has a lower slope.

8. Expected return for equity fund = T-bill rate + Risk premium = 6% + 10% = 16%
   Expected rate of return of the client's portfolio = (0.6 × 16%) + (0.4 × 6%) = 12%

   Expected return of the client's portfolio = 0.12 × \$100,000 = \$12,000

   (which implies expected total wealth at the end of the period = \$112,000)

   Standard deviation of client's overall portfolio = 0.6 × 14% = 8.4%

9. Reward-to-volatility ratio $= \dfrac{0.10}{0.14} = 0.71$